Menoufia University		**Mid-Term Term Exam**
Faculty of Electronic Engineering		**Date: 21 / 11 / 2023**
Program: All Programs		**No. of Exam Pages: 1**
Course Title: Electrical Measurements		**No. of Exam Questions: 2**
Academic Level: 2 / Semester: 1		**Exam Marks: 20 Marks**
Academic Year: 2022 / 2023		**Exam Time: 1 Hour**
Course Code: ACE262		**From 11:00 AM to 12:00 AM**

Answer the following questions:

Question – 1:

Put True (√) or False (×) signs for the following expressions (with Correction if found): [10 Marks]

1. **Threshold is a lower limit on the magnitude of the change in the input measured quantity that produces an observable change in the instrument output.**
2. **A car speedometer typically has a resolution of about 15 km/h. This means that, if the vehicle starts from rest and accelerates, no output reading is observed on the speedometer until the speed reaches 15 km/h.**
3. **In the active instrument the external power supply is not need.**
4. **Measurement system is useful in feedback control systems only.**
5. **An analog instrument gives an output that varies continuously as quantity being measured.**
6. **The threshold of an instrument defines the minimum and maximum values of a quantity that the instrument is designed to measure.**
7. **Tolerance is a term that describes an instrument's degree of freedom from random errors.**
8. **The precision of an instrument is a measure of how close the output reading of the instrument is to the correct value.**
9. **Reproducibility describes the closeness of output readings when the same input is applied repetitively over a short period of time, with the same measurement conditions.**
10. **If a large number of readings are taken of the same quantity by a high precision instrument, then the spread of readings will be very small.**

Question – 2: **[10 Marks]**

1. **Draw the block diagram of measurement system. [2.5 marks]**
2. **State the disadvantages of the conversion from A/D. [2.5 marks]**
3. **A voltmeter, having a sensitivity of 1,000Ω /V, reads 100 V on its 150-V scale when connected across an unknown resistor in series with a millimeter. When the millimeter reads 5 mA, Calculate:**
   1. Apparent resistance of the unknown resistor,
   2. Actual resistance of the unknown resistor,
   3. Error due to the loading effect of the voltmeter. [5 marks]

With best wishes

Dr. Mohamed I.Abdo